CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 26-2021: A 49-Year-Old Man with Relapsed Acute Myeloid Leukemia

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PRESENTATION OF CASE

Dr. Richard A. Newcomb: A 49-year-old man was evaluated at this hospital because of relapsed acute myeloid leukemia (AML).

Fourteen months before the current presentation, the patient was evaluated at another hospital because of pain in the right upper quadrant that had begun 4 days earlier and was associated with nausea and vomiting. Abdominal imaging revealed evidence of acute cholecystitis. Initial laboratory test results showed peripheral blasts that were suggestive of acute leukemia. The patient was transferred to a second hospital for additional evaluation. At the second hospital, abdominal imaging confirmed acute cholecystitis, and a calculus was found lodged in the neck of the gallbladder. The white-cell count was 4200 per microliter (reference range, 4200 to 9900), of which 40% were blasts. A peripheral-blood smear was notable for circulating blasts with Auer rods. The hemoglobin level was 11.7 g per deciliter (reference range, 13.0 to 17.4), and the platelet count was 22,000 per microliter (reference range, 140,000 to 440,000). A bone marrow biopsy was performed, followed by laparoscopic cholecystectomy.

Dr. Andrew M. Crabbe: The bone marrow biopsy performed at the second hospital was inadequate owing to the small size of the specimen; however, intact bone marrow particles showed a cellularity of 95 to 100%. A 300-cell differential count on combined touch-imprint and aspirate smears showed 39% blasts, 3% promyelocytes, 19% neutrophils and precursors, 5% erythroid precursors, 30% lymphocytes, 3% monocytes, 1% eosinophils, and less than 1% basophils. The blasts were medium in size, with irregular nuclear contours, moderately clumped chromatin, occasional prominent small nucleoli, and scant granular cytoplasm. Auer rods were identified. Flow cytometry of the peripheral blood identified myeloid blasts that were negative for CD34 and HLA-DR and positive for CD117, CD33, CD13, and myeloperoxidase. Cytogenetic analysis of the peripheral blood with the use of fluorescence in situ hybridization (FISH) revealed a normal karyotype (46,XY) and no evidence of a t(15;17) chromosomal translocation. Molecular testing of the peripheral blood by means of dedicated polymerase-chain-reaction (PCR) testing

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and next-generation sequencing identified mutations in *NPM1*, *DNMT3A*, and *NRAS* and did not identify mutations in *FLT3*, *IDH1*, or *IDH2*. The overall findings of this testing were consistent with the World Health Organization classification of AML with mutated *NPM1*.

Dr. Newcomb: Induction chemotherapy with 7 days of cytarabine and 3 days of daunorubicin was administered. On day 34 after the initiation of chemotherapy, the absolute neutrophil count was 2.2 per microliter (reference range, 1.5 to 7.5), and the platelet count was 536,000 per microliter. A bone marrow biopsy performed on day 37 after initiation of chemotherapy revealed hypercellular marrow with maturing trilineage hematopoiesis. There was no morphologic or phenotypic evidence of persistent or relapsed AML. A 300-cell sample of the bone marrow aspirate contained less than 1% blasts. Consolidation chemotherapy with four cycles of high-dose cytarabine was completed 8 months before the current presentation.

Dr. Crabbe: A bone marrow biopsy performed after the final cycle of chemotherapy revealed normocellular marrow with trilineage hematopoiesis; there was no morphologic or phenotypic evidence of AML. Next-generation sequencing of the bone marrow biopsy specimen revealed no pathogenic mutations.

Dr. Newcomb: The patient remained well for 7 months. One month before the current presentation, mild leukopenia developed, and more frequent laboratory monitoring was instituted. One week before the current presentation, the white-cell count was 2600 per microliter (reference range, 4000 to 10,000), of which 13% were peripheral blasts. The patient was referred to this hospital for evaluation.

On presentation to this hospital, the patient reported a 3-week history of progressive fatigue, dyspnea on exertion, exercise intolerance, widespread muscle aches, and night sweats. He noted that similar symptoms had preceded his initial diagnosis.

Medical history included obesity and hypertension. The patient was not taking medications on a regular basis. He was allergic to chlorhexidine, which caused rash. He lived in rural New England with his wife and four children. He worked as a software engineer. He did not smoke, and he drank alcohol rarely. His mother had coronary artery disease, and a sister had had a stroke. He had three other siblings who were healthy. There was no family history of leukemia or other hematologic disorders.

On examination, the patient appeared well. The temperature was 35.7°C. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 43. No lymphadenopathy, rashes, or skin lesions were present. The remainder of the examination was normal. The white-cell count was 5830 per microliter (reference range, 4500 to 11,000), of which

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, This Hospital†	On Presentation, This Hospital
Hemoglobin (g/dl)	12.0–16.0	14.6
Hematocrit (%)	41.0–53.0	43.2
White-cell count (per μl)	4500–11,000	5830
Differential count (%)		
Neutrophils	40–70	12.6
Lymphocytes	22–44	39.7
Reactive lymphocytes	0	2.7
Eosinophils	0–8	0.9
Basophils	0–3	0.9
Monocytes	0–11	1.8
Blasts	0	39.6
Platelet count (per μl)	150,000–450,000	44,000
Creatinine (mg/dl)	0.60–1.50	1.00
Urea nitrogen (mg/dl)	8–25	11
Aspartate aminotransferase (U/liter)	9–32	25
Alanine aminotransferase (U/liter)	7–33	21
Alkaline phosphatase (U/liter)	30–100	97
Total bilirubin (mg/dl)	0.0–1.0	0.1
Albumin (g/dl)	3.3–5.0	4.0
Globulin (g/dl)	1.9–4.1	2.9
Lactate dehydrogenase (U/liter)	110–210	459
Uric acid (mg/dl)	3.6–8.5	5.5
Glycated hemoglobin (%)	4.3–6.4	6.0

* To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for uric acid to micromoles per liter, multiply by 59.48.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

39.6% were blasts. Additional laboratory test results are shown in Table 1. A bone marrow biopsy was performed, and management decisions were made.

DISCUSSION OF BONE MARROW BIOPSY RESULTS

Dr. Crabbe: The bone marrow biopsy revealed hypercellular marrow containing sheets of immature cells that were consistent with myeloid blasts (Fig. 1A and 1B). Examination of the bone marrow aspirate showed medium-sized blasts with scant sparsely granular cytoplasm (Fig. 1C); the blasts accounted for 89% of the marrow cellularity. On flow cytometry of the aspirate, the blasts were aberrantly negative for CD34, CD117, and HLA-DR and were mostly positive for myeloperoxidase. Although this immunophenotype was atypical, it was similar to that seen at initial diagnosis. These findings were consistent with relapsed AML.

A comparison of the molecular genetic profiles at initial diagnosis, at 6-month follow-up, and at relapse are shown in Table 2. At initial diagnosis, the patient had a normal karyotype with mutations in *NPM1*, *DNMT3A*, and *NRAS* identified on targeted next-generation sequencing. However, at relapse, the mutations in *NPM1* and *NRAS* were no longer detected, and a new internal tandem duplication mutation in *FLT3* (*FLT3*-ITD) was identified by a PCR-based sizing assay. In addition, t(3;9) and t(3;19) translocations were identified on karyotype analysis, although these translocations did not correspond to any reportable fusion transcripts on a next-generation sequencing fusion assay, and FISH studies for rearrangements involving *MECOM* (located on chromosome 3) were negative. Although these translocations are of uncertain clinical significance, they serve as markers of clonal evolution.

It is not typical for patients who have AML with mutated *NPM1* to have a relapse with wild-type *NPM1*; however, such a finding has been observed in 1 to 14% of relapses in various studies.[1-3] A prevailing theory is that relapses without a detectable mutation in *NPM1* arise from preleukemic clones that were present at diagnosis. The high frequency of persistent *DNMT3A* mutations in these cases[1,3] supports this theory. The acquisition of mutations in certain genes, such as *DNMT3A*, *TET2*, or *ASXL1*, is a well-recognized early or preleukemic event in the clonal evolution to AML. For this patient, the same *DNMT3A*

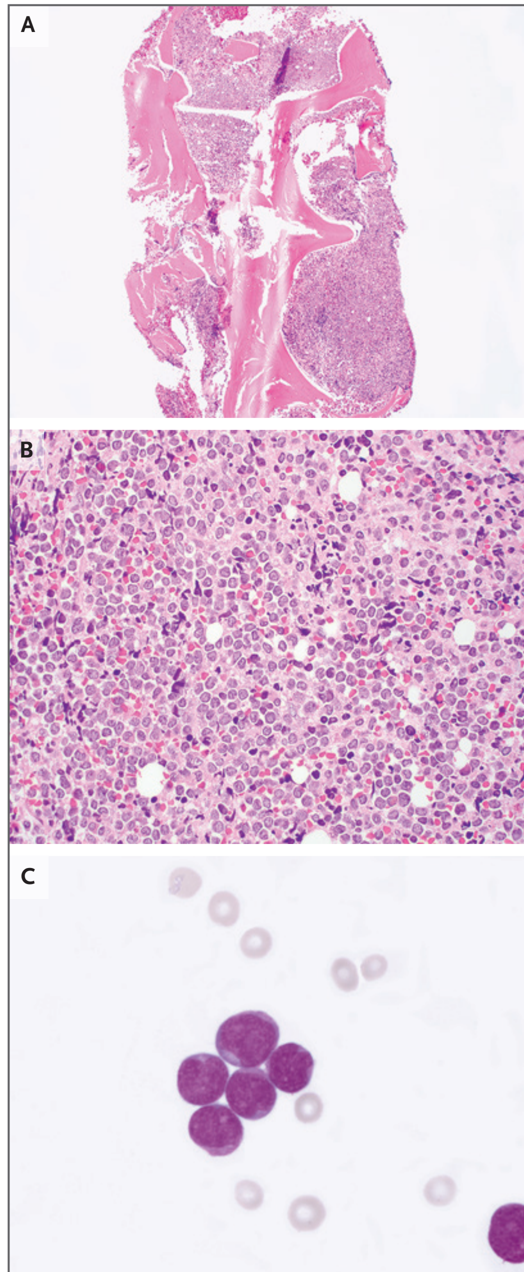


Figure 1. Bone Marrow Biopsy Specimen and Aspirate.
Hematoxylin and eosin staining of a biopsy specimen (Panel A, and at higher magnification in Panel B) shows markedly hypercellular marrow containing immature cells that are consistent with myeloid blasts. Wright–Giemsa staining of the aspirate (Panel C) shows medium-sized myeloid blasts with slight nuclear irregularities and scant sparsely granular cytoplasm.

Table 2. Results of Genetic Testing of Bone Marrow.*

Assessment	At Initial Diagnosis, 14 Mo before Presentation	At 6-Mo Follow-up, 8 Mo before Presentation	At Relapse, 1 Mo before Presentation
Histologic diagnosis	AML	Negative for AML	AML
Cytogenetic analysis	46,XY[20]	46,XY[20]	46,XY, t(3;9) (q29;p24), t(3;19) (q29;q13.3)[20]
SNaPshot NGS assay	*DNMT3A* splice variant c.1015–2A→G (VAF, 29%) *NPM1* p.Ile269Alafs*7 (VAF, 20%) *NRAS* p.Gly13Asp (VAF, 23%)	Negative	*DNMT3A* splice variant c.1015–2A→G (VAF, 41%)
NGS fusion assay	Negative	Not performed	Negative
FLT3 sizing assay	Negative	Not performed	Positive for *FLT3*-ITD variant

* "[20]" denotes that 20 metaphases were examined for karyotypic analysis. AML denotes acute myeloid leukemia, *FLT3*-ITD internal tandem duplication mutation in *FLT3*, NGS next-generation sequencing, and VAF variant allele frequency.

splice acceptor variant was present at both the time of initial diagnosis and the time of relapse; this pattern is consistent with that observed in these similar cases.

PATHOLOGICAL DIAGNOSIS

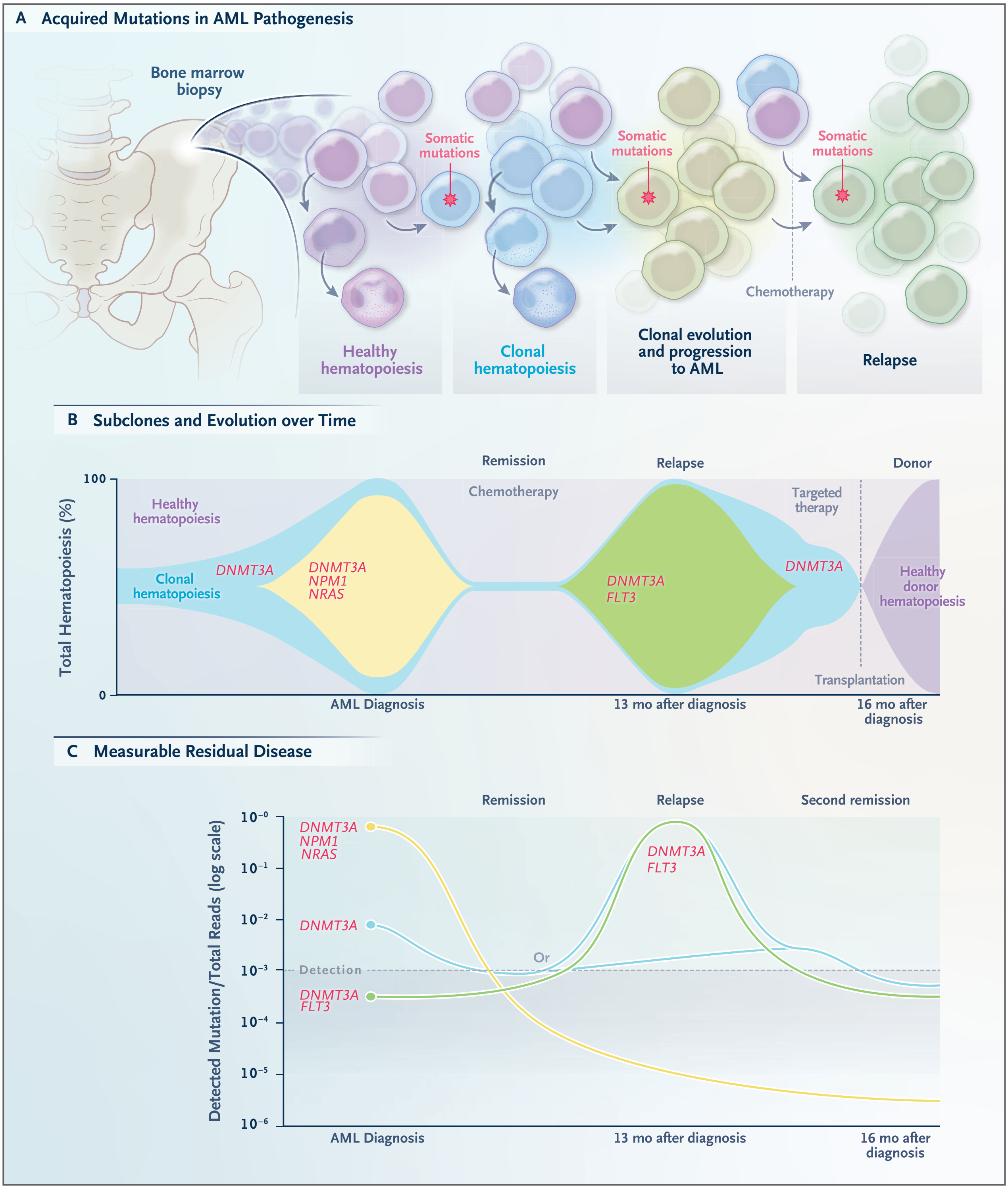
Relapsed acute myeloid leukemia.

DISCUSSION OF MANAGEMENT

Dr. Andrew M. Brunner: This 49-year-old man had a subtype of AML that had arisen de novo, with a normal male karyotype and mutations in *NPM1*, *DNMT3A*, and *NRAS* that had been detected at the time of initial diagnosis of AML, 14 months before this presentation. Given these molecular features, this patient was considered to have favorable-risk disease and was treated with induction and consolidation chemotherapy, and within a year after treatment, he had a relapse of AML — this time with an altered mutational profile. This case illustrates our growing understanding of the clonal heterogeneity of AML and its evolution over time with therapy. Such clonal diversity has led to new therapeutic approaches that have resulted in treatment that is increasingly tailored to each patient.

Recurrent somatic cytogenetic and molecular alterations inform the initial prognosis of patients with AML, and during the last 15 years, the mutational landscape of AML has been greatly refined.[4-8] At diagnosis, this patient underwent conventional risk stratification through the characterization of cellular features, including through morphologic evaluation, cytogenetic analysis, and evaluation for a limited number of molecular alterations (Fig. 2).[9] The selection of initial therapy in adults with newly diagnosed AML is often largely based on patient "fitness," which may be assessed in numerous ways, including on the basis of age and coexisting conditions,[10] functional assessments,[11] and patient preference. Younger, more "fit" patients are usually treated with an intensive cytarabine-based induction regimen, whereas patients who are less fit may receive lower-intensity chemotherapy (e.g., a hypomethylating agent such as azacitidine or decitabine, with or without the addition of venetoclax, or treatment with low-dose cytarabine), or supportive care alone. Assessment of fitness can be subjective, and there are concerns that it could lead to undertreatment of patients at the time of AML diagnosis; population-based assessments of treatment patterns in AML suggest that many older patients do not receive any chemotherapy,[12] which poses a challenge, given that the median age at AML diagnosis is nearly 70 years.

Given this patient's younger age (<60 years) and overall good health at the time that de novo AML was diagnosed, he received induction chemotherapy with the combination of 7 days of continuous cytarabine and 3 days of anthracycline, or "7 plus 3" chemotherapy. Had the patient received the initial diagnosis of AML today, it is likely that — if his condition were clinically stable — decisions regarding the selection of induction treatment would incorporate the results of evaluations for specific cytogenetic and molecular alterations. For instance, on the basis of the identification of an *FLT3* mutation, we



would add the FLT3-directed tyrosine kinase inhibitor midostaurin to treat *FLT3*-mutated AML.[13] Alternatively, we would add gemtuzumab–ozogamicin as part of the treatment of core-binding-factor–mutated AML,[14] or in the case of AML harboring a mutation in *TP53*, we would consider treatment with a hypomethylating agent as the therapeutic backbone.[15] This patient did

Figure 2 (facing page). Somatic Mutations, Clonal Hematopoiesis, and Their Relation to Acute Myeloid Leukemia (AML).

Hematopoietic progenitors in the bone marrow can acquire somatic mutations, which may result in clonal expansion of immature myeloid blasts over time or in the context of environmental exposure (Panel A). AML can develop from specific acquired mutations in an otherwise healthy progenitor, or it may result from background clonal hematopoiesis. Plotting the proportion of cells with shared clonal characteristics can show the dynamic evolution to initial diagnosis of AML, illustrated here with a precursor *DNMT3A* mutation and with subsequently acquired *NPM1* and *NRAS* mutations (Panel B). At remission, there is a small residual *DNMT3A* clone below the level of detection of standard clinical next-generation sequencing that subsequently expands with a concurrent *FLT3* mutation. After hematopoietic stem-cell transplantation, the marrow is replaced by donor hematopoiesis, which is typically monitored by means of chimerism studies. Measurable residual disease can be tracked over time and according to clonal characteristics. In this patient, the predominant clone at diagnosis harbored mutations in *DNMT3A*, *NPM1*, and *NRAS*; after treatment, this clone diminished, but at relapse, a clone with mutations in *DNMT3A* and *FLT3* was present. The clone present at relapse arose either from a residual pretreatment subclone (Panel C, green line) or as a result of a newly acquired *FLT3* mutation (blue line). The depth of a molecular response may predict overall outcomes, including the post-transplantation risk of relapse.

not harbor a mutation that would have necessarily changed the initial therapy; his early relapse, within a year after completing consolidation chemotherapy, offers an opportunity to discuss changes in how we monitor AML and tailor subsequent therapy.

DISEASE MONITORING IN CLINICAL PRACTICE

Our current understanding of the pathogenesis of AML is that cancerous cells arise from an early hematopoietic progenitor through the acquisition of somatic mutations, which leads to clonal expansion of immature myeloid blasts and a malignant phenotype. The mutations leading to this presentation vary; some are more directly leukemogenic, such as the t(15;17) *PML-RARA* translocation,[16] whereas others require additional cooperating mutations to progress, such as cases in which mutations in *DNMT3A*, *NPM1*, and *FLT3* are present.[17] These mutations may all arise within the same originating clone, or they may be acquired in various subclones with differing leukemogenic potential until a dominant clone emerges, resulting in the leukemic phenotype.[5] The mutation profile that is subsequently detected in a blood or bone marrow sample is typically a reflection of the most abundant subclone, and the subclone that is the most abundant may evolve over time.

Somatic mutations that are specific to leukemic cells provide a unique opportunity, not only for risk stratification and targeted therapeutics but also for distinguishing these cells from healthy hematopoietic progenitors. If such mutations are truly specific to the leukemia and can be measured with sufficient sensitivity, they can serve as a measure of depth of remission and can be tracked and monitored during therapy.[18] There are several well-established examples of this strategy, such as the monitoring of BCR-ABL fusion protein levels in patients with chronic myeloid leukemia.

However, there have been challenges with this approach in monitoring AML. First, the mutational heterogeneity lies not only in the genes involved but also in the specific alteration present (e.g., missense, nonsense, frameshift, and insertion or deletion lesions, which are distinct from variants of undetermined significance).[19] Second, cooperating mutations can vary over time, or subclones that were not readily detected at diagnosis may expand.[16]

Further complicating the assessment of measurable residual disease is the increasing frequency of detectable clonal hematopoiesis with advancing age, a finding seen in a number of otherwise healthy persons.[20,21] We often face the clinical uncertainty of a patient who is in remission but still has some of the mutations that were present at diagnosis. Although some mutations are clearly part of the leukemic clone, we may not know whether some of the other persistent mutations merely indicate underlying clonal hematopoiesis, particularly among older adults with AML in whom clonal hematopoiesis is otherwise common. Some studies suggest that residual mutations that frequently characterize clonal hematopoiesis — mutations in *DNMT3A*, *TET2*, or *ASXL1*, so-called DTA mutations — may not be predictive of AML relapse after chemotherapy and therefore are less likely to be associated with the leukemogenic clone.[22]

NPM1 MUTATION

These limitations aside, one of the mutations present in this patient's diagnostic peripheral-blood sample has been studied extensively with respect to measurable residual disease assessment: the mutation in *NPM1*. There are several factors that make disease monitoring based on *NPM1* alterations appealing. For example, this mutation seems to be relatively specific to the leukemic clone, and evidence suggests that it is stable, meaning that most patients with *NPM1*-mutated AML who subsequently have a relapse will still have that mutation after relapse. In addition, unlike many other recurrently mutated genes in AML, more than 90% of the pathogenic mutations in *NPM1* are an insertion of four base pairs, which facilitates the development of assays for tracking these mutations at low levels. This patient had a concurrent mutation in *DNMT3A*, but distinguishing this mutation from clonal hematopoiesis can be challenging; he also had an *NRAS* mutation, which often occurs later during clonal evolution and is more likely than other mutations to be lost during treatment.[17]

If we apply these principles to this patient, we note that after his course of chemotherapy, a bone marrow assessment to evaluate morphologic remission was performed, as well as a clinical next-generation sequencing–based assay, which did not detect the *NPM1* mutation. The presence of detectable disease after induction chemotherapy or before allogeneic hematopoietic stem-cell transplantation has been associated with higher rates of relapse and worsened survival. In this case, the clinical next-generation sequencing–based assay that was performed would generally be able to detect an *NPM1* mutation in approximately 1 to 5 of every 100 reads, which may be below the level of morphologic detection but not as sensitive as some other assays that have been developed for the assessment of *NPM1* measurable residual disease. There is substantial debate about which assay to use, the threshold for what constitutes measurable residual disease negativity, and interventions that may be used on the basis of such findings, all of which highlight the need for prospective studies.[22-25]

Moreover, although the *NPM1* mutation is detectable at relapse in most patients with *NPM1*-mutant AML who have a relapse, a small percentage of patients (with estimates ranging from 1 to 14%) will no longer have a detectable *NPM1* mutation. One series showed that the majority of such patients had concurrent *DNMT3A* mutations,[1] as was seen in this case. This observation raises the question of whether tracking *NPM1* mutation levels alone is sufficient in all patients with *NPM1*-mutated AML; this patient may have had eradication of the *NPM1* component of his AML, but residual *DNMT3A*-mutated disease was present at relapse. In one study in which *NPM1* measurable residual disease was used for tracking disease, the risk of relapse in the subgroup of patients with a concurrent *DNMT3A* mutation at diagnosis was nearly 50%, despite the achievement of *NPM1* measurable residual disease negativity.[2] Even if an *NPM1* measurable residual disease assay had been used to monitor this patient during treatment, it may have been falsely reassuring, given the mechanism of relapse that was observed here. Such cases also raise the question of how to interpret possible clonal hematopoiesis; for instance, not all DTA mutations may be safe to "ignore." Ongoing efforts to study measurable residual disease monitoring frequently involve the use of multiple tools, including leukemia-specific flow cytometry and single-cell genetic sequencing. Such assays are starting to be incorporated into patient care; however, testing that is currently used in clinical practice can vary with respect to its sensitivity and specificity for measuring residual disease.

MANAGEMENT OF RELAPSED AML

In patients with relapsed AML, such as this patient, there are three general questions that I seek to answer: What is the clinical condition of the patient? What is the molecular and cytogenetic profile and behavior of the AML at relapse? What are the goals of therapy? Historically, this patient would have been considered for intensive reinduction chemotherapy if his overall health and performance status remained adequate, or he would have been considered for lower-intensity chemotherapy if his status were inadequate. However, more recently, substantial changes in the approach to AML therapy — at diagnosis and at relapse — have led to new strategies for the management of relapsed AML.

It is increasingly important to recharacterize the AML mutational profile at therapeutic decision points, particularly at relapse. As noted, at relapse, this patient had a stable *DNMT3A* mutation, but a new subclone harboring a concurrent *FLT3*-ITD mutation had emerged. *FLT3* alterations represent potential targets for effective therapies; it is also not uncommon for these mutations to be variably detected during treatment, as the AML clonal structure evolves.

When a discussion with any patient with relapsed AML is initiated, particularly in cases in which expanded therapeutic options are being considered, it is important to clarify the overall goals of therapy. At diagnosis, the goal was to administer chemotherapy with curative intent. However, with relatively early disease relapse, the best chance of cure will involve allogeneic hematopoietic stem-cell transplantation. Therefore, my goal for this patient would be to select a regimen that would provide the best chance of a remission and ideally would achieve a state of very low measurable residual disease, which has been shown to predict post-transplantation outcomes.[26,27]

At the time of relapse, some patients have coexisting conditions or complications of treatment or disease, or their primary goal is disease palliation. In such cases, the assessment of risk and benefit may differ — a prolonged hospitalization for chemotherapy may not be aligned with a patient's values. That being said, there can be a clinically significant burden of symptoms associated with AML, and even if treatment is given only for palliative intent, therapies that result in few side effects and that can control disease and improve blood-cell counts can yield meaningful patient outcomes. It is also important to remember that disease response and clinical fitness can change with time, so even if a patient is not "ready" clinically to undergo transplantation at the time of relapse, this status can change as the disease is treated.

For this patient with relapsed or refractory *FLT3*-mutated AML, the oral FLT3-directed tyrosine kinase inhibitor gilteritinib is considered to be standard care.[28] Patients who receive gilteritinib have a better response and longer overall survival than those who receive standard salvage chemotherapy. Still, the probability of achieving a complete remission with single-agent gilteritinib is relatively low, at 21%, although 54% of patients have a bone marrow response such that allogeneic hematopoietic stem-cell transplantation could be considered.

DYNAMIC ASSESSMENT AS A STRATEGY IN AML

As recently as 5 years ago, the general approach to this patient would have followed a different framework. Although cytogenetic and molecular risk would have been assessed, these would have been used primarily to decide whether to proceed with transplantation or to administer consolidation chemotherapy in first remission. This patient had favorable disease features and thus would have proceeded with consolidation chemotherapy (as he did), and when he had a relapse, treatment would have been selected largely on the basis of clinical fitness, with consideration of either intensive salvage options (with a goal of proceeding to transplantation) or palliative chemotherapy options.

Now, our understanding of the dynamic clonal features of AML throughout treatment plays a more substantial role in treatment decisions. Along with the use of serial disease measurement in informing targeted therapies for initial treatment, there is a growing literature to support the use of such a strategy to inform and modify subsequent treatment selection. Clearance of measurable residual disease is increasingly being studied as a way of further tailoring care, in addition to standard disease risk stratification; for example, such an assessment can help to identify patients who may benefit from treatment intensification and transplantation.[24,27] The ability to track low levels of disease and to understand the way in which subclonal disease can persist after therapy have also prompted new areas of clinical research, such as the evaluation of maintenance therapies to prevent disease relapse in high-risk patients.[29,30] However, the underlying features of these efforts stem from our understanding of somatic mutations with respect to their contribution to AML pathogenesis and their variation over time, as well as our technical ability to detect and follow leukemia-specific variants, a discipline that itself continues to evolve.

Dr. Yi-Bin Chen: When this patient was referred to us, we asked his local oncologist to delay initiation of planned conventional salvage chemotherapy so that we could perform additional diagnostic testing, specifically mutational analysis. Given the relatively short duration of remission between completion of conventional high-dose cytarabine consolidation chemotherapy and identification of the new *FLT3*-ITD mutation, we recommended proceeding with a more targeted therapeutic approach that consisted of the combination of azacitidine and gilteritinib.

The patient's clinical course was complicated by the development of hepatosplenic candidiasis during the first cycle of therapy and the lack of meaningful improvement in the blood-cell count, which persisted through the end of the second cycle. He returned to this hospital for a restaging bone marrow biopsy; examination of the bone marrow specimen revealed hypocellular marrow but no morphologic evidence of AML. A next-generation sequencing–based *FLT3*-ITD measurable residual disease assay showed the presence of measurable residual disease at the level of 3.94×10^{-2} mutant–wild-type variant read frequency. At this juncture, we did not think that additional chemotherapy would be beneficial, given the patient's achieved response, lack of hematopoietic recovery, and potential for infectious complications.

The patient proceeded expeditiously to allogeneic hematopoietic stem-cell transplantation, receiving HLA-matched peripheral-blood progenitor cells from his brother. His conditioning regimen for transplantation included fludarabine and melphalan. His post-transplantation course was uncomplicated, with engraftment of donor counts by day 14. Given his *FLT3*-ITD mutation, gilteritinib was reinitiated approximately 1 month after hematopoietic stem-cell transplantation to decrease the risk of relapse. The use of FLT3 inhibitors as maintenance therapy after hematopoietic stem-cell transplantation in patients with *FLT3*-ITD–mutated AML in first remission has been shown to be beneficial,[31-34] and we await results from an international phase 3 trial in which gilteritinib is used as maintenance therapy (ClinicalTrials.gov number, NCT02997202). Currently, 3 months after hematopoietic stem-cell transplantation, the patient is well; he has had a full recovery, with no evidence of graft-versus-host disease. A restaging bone marrow biopsy that will include assessment of *FLT3* measurable residual disease has been scheduled, and tapering of his immunosuppressive therapy has begun.

FINAL DIAGNOSIS

Relapsed acute myeloid leukemia (with wild-type *NPM1* and newly identified internal tandem duplication mutation in *FLT3*).

This case was presented at Cancer Center Grand Rounds.
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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